Akshat Kalra

4th Year Computer Science & Statistics | UBC

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Education

The University of British Columbia — Vancouver, BC

BSc Computer Science and Statistics | ***Dean's Honor List*** — *Expected Graduation May 2027*

- **Relevant Coursework:** Statistical Modelling (95%), Probability Theory (91%), Computer Networking (97%), Software Construction (91%), Models of Computation (93%).
- **Scholarships:** 2x Faculty of Science International Student Scholarship (2025, 2024), OIS Scholarship (2022).
- **Leadership:** Notion Campus Leader, VP Internal - Brock Commons Residence Association.

Experience

Software Developer (GenAI) Co-op — May 2025 – April 2026

Faculty of Forestry, UBC | ***TypeScript, Next.js, Nest.js, Python, AWS*** — *Vancouver, BC*

Project: ***AI-Assisted Grading Helper*** | *Cloud Infrastructure, LTI Integration, Model Evaluation*

- Owning the full development cycle of the AI assisted grading helper tool, architecting **AWS serverless infrastructure** (**Lambda, DynamoDB, API Gateway**) provisioned via **SAM/CloudFormation** with automated deployment through **GitHub Actions CI/CD**.
- Implemented **LTI 1.3** Canvas integration with automatic grade passback and secure **OAuth 2.0** authentication for seamless instructor workflows.
- Implemented an LLM grading evaluation system following the methodology of **Impey et al. (2025)**, benchmarking Claude, Llama 3, and Mistral via **AWS Bedrock** against instructor grades on **3,549 student submissions** across 81 questions using ICC, RMS metrics, and bootstrap confidence intervals for statistical validation.

Project: ***HelpMe*** | *Open-Source LLM-Powered Student Course Assistant*

- Worked with **Prof. Ramon Lawrence's** research team to design and develop production features for **HelpMe**, an open-source LLM powered course assistant deployed across **40+ courses** and used by **1,500+ students**.
- Engineered a data pipeline ingesting Canvas LMS course content into vector embeddings using **pgvector (PostgreSQL)**, enabling semantic search for the **Retrieval Augmented Generation** pipeline.
- Designed and implemented **NestJS REST API** endpoints and **TypeORM-PostgreSQL** schemas to synchronize Canvas course data and manage embedding storage, enabling fast and reliable semantic search.
- Built instructor-controlled backend filtering to restrict LLM retrieval by resource type, visibility, and metadata, preventing exposure of hidden quizzes and answer keys; contributed integration and unit tests to the open-source production codebase.

Undergraduate Teaching Assistant — Sep 2023 – Present

Department of Computer Science & Philosophy, UBC — *Vancouver, BC*

CPSC 121 (Models of Computation) | 4 Terms: 2024W2, 2025S2, 2025W1, 2025W2

- Led weekly labs, tutorials, and discussion sections teaching automata theory, boolean logic, proof techniques, and sequential circuit design.
- Developed original discussion slides relating problem sets to lecture content; held structured office hours promoting independent problem-solving and active student engagement.

PHIL 220 (Symbolic Logic) | 2 Terms: 2023W1, 2024W1

- Graded 200+ assignments and exams (midterms and finals); provided detailed written feedback on sentential logic, predicate logic, and formal proof construction.
- Assisted in writing new exam and bonus questions; held weekly office hours and maintained Piazza discussion forum to support student learning.

Machine Learning Projects

Airbnb Policy Prediction | ***1st Place at Case Competition*** *(Won $300)* | *Python, Scikit-learn* Nov 2024

- Built end-to-end ML pipeline using Random Forest regressor that **won $300 and first place** at the competition.
- Conducted feature engineering, hyperparameter tuning (RandomizedSearchCV) and recursive feature elimination, reducing MSE from 1.64M to 1.19M.
- Conducted exploratory data analysis; documented methodology and insights in Medium article.

Restaurant Revenue Inferential Analysis | R, tidyverse, ggplot2, Poisson Regression Apr 2025

- Analyzed 1,000 synthetic restaurant records using quasi-Poisson regression to address overdispersion; identified Monthly Revenue as strongest predictor of customer footfall and Promotions as significant positive factor.
- Conducted exploratory data analysis with ggplot2 correlation matrices, performed hypothesis testing on 7 predictor variables, and validated model fit through deviance residual diagnostics and AIC comparison to ensure statistical reliability.

Eco-Circle | ***UBC CIC × AWS Hackathon 3rd Place*** *(Won $200)* | AWS, Next.js, Python Oct 2024

- Developed an AI-powered sustainable marketplace that connects eco-conscious buyers and sellers through personalized upcycling, reuse, and recycling recommendations generated via Amazon Bedrock (Llama 3.1 70B).
- Architected serverless infrastructure using AWS Lambda, DynamoDB, S3, and AWS Bedrock for LLM inference.

Software Engineering Projects

Rustlock | Rust, AES-256-GCM, Argon2id, Clap Feb 2025

- Built a local-first **CLI password manager in Rust** with AES-256-GCM authenticated encryption and Argon2id key derivation configured with 64MB memory cost, providing resistance against GPU-based brute force attacks.
- Implemented cryptographically secure password generation using OS-level entropy across a 92-character alphabet, encrypted vault persistence with salt/nonce prepending, and modular architecture across 6 components.

TCP Implementation | C, Linux Socket APIs, Network Protocols Mar 2025

- Implemented a TCP-like protocol over UDP with three-way handshake, sliding window flow control, and exponential backoff retransmission (1-4s) to ensure reliable data transfer under network adversity.
- Implemented fast retransmit on 3 duplicate ACKs, IP checksum validation, and 32-bit sequence number wraparound handling.

Technical Extracurriculars

Hackathons & Case Competitions

- UBC WiDS Case Competition **(Winner)**, UBC GDSC × Launchpad Hackathon **(Winner)**, UBC CIC × AWS GenAI & Sustainability Hackathon **(3rd Place)**, nwPlus cmd-f 2025 **(Mentor & Judge)**, nwPlus HackCamp 2025 & 2024 **(Mentor)**, ASA Datafest 2025 **(Mentor)**

Technical Skills

Languages: Python, C/C++, TypeScript, Java, Rust, SQL (PostgreSQL), R, HTML/CSS, Racket (Lisp Dialect)
Machine Learning & AI: Scikit-Learn, Neural Networks (autograd from scratch), PyTorch, Automated Evaluation for AI systems, AWS Bedrock
Data Science & Analytics: Pandas, NumPy, Matplotlib, Statistical Validation
Cloud & Infrastructure: AWS (Bedrock, Lambda, S3, DynamoDB, API Gateway), Docker, Linux Operating System
Full-Stack: React.js, Next.js, Nest.js, Node.js, FastAPI, Flask